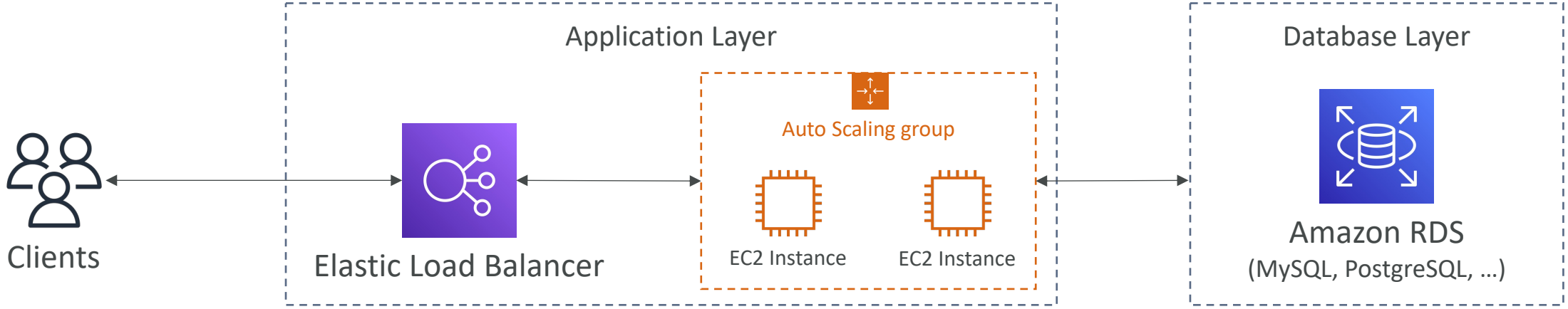
Amazon DynamoDB

NoSQL Serverless Database

Traditional Architecture



- Traditional applications leverage RDBMS databases
- These databases have the SQL query language
- Strong requirements about how the data should be modeled
- Ability to do query joins, aggregations, complex computations
- Vertical scaling (getting a more powerful CPU / RAM / IO)
- Horizontal scaling (increasing reading capability by adding EC2 / RDS Read Replicas)

NoSQL databases

- NoSQL databases are non-relational databases and are **distributed**
- NoSQL databases include MongoDB, DynamoDB, …
- NoSQL databases do not support query joins (or just limited support)
- All the data that is needed for a query is present in one row
- NoSQL databases don't perform aggregations such as "SUM", "AVG", …
- **NoSQL databases scale horizontally**

- There's no "right or wrong" for NoSQL vs SQL, they just require to model the data differently and think about user queries differently

Amazon DynamoDB

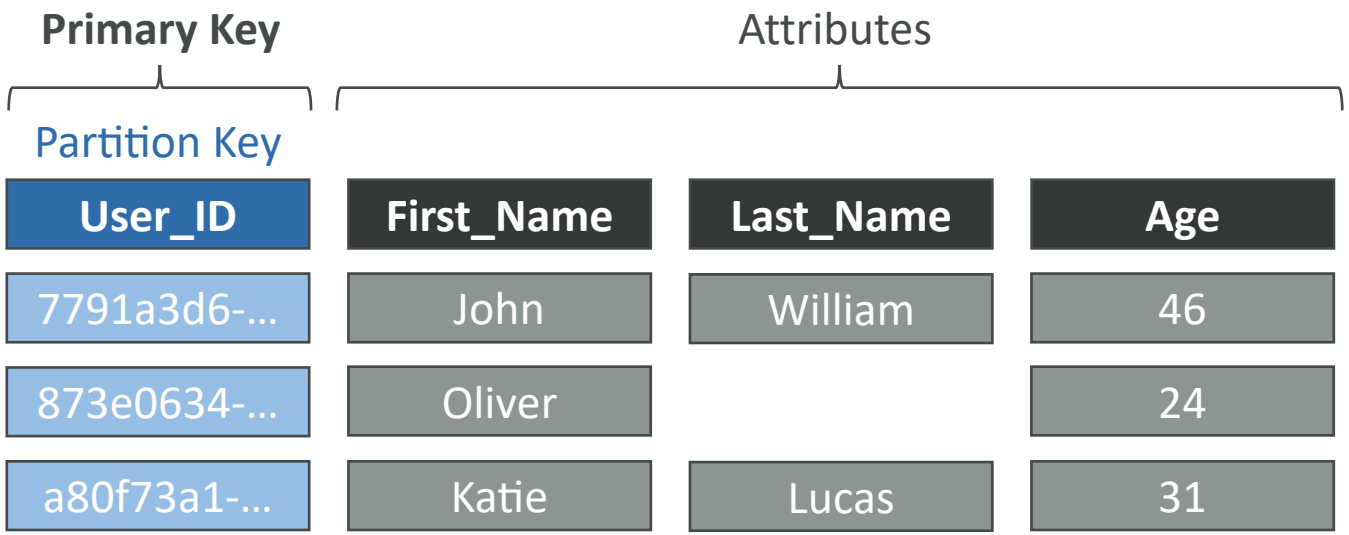
- Fully managed, highly available with replication across multiple AZs
- NoSQL database - not a relational database
- Scales to massive workloads, distributed database
- Millions of requests per seconds, trillions of row, 100s of TB of storage
- Fast and consistent in performance (low latency on retrieval)
- Integrated with IAM for security, authorization and administration
- Enables event driven programming with DynamoDB Streams
- Low cost and auto-scaling capabilities
- Standard & Infrequent Access (IA) Table Class

DynamoDB - Basics

- DynamoDB is made of **Tables**
- Each table has a **Primary Key** (must be decided at creation time)
- Each table can have an infinite number of items (= rows)
- Each item has **attributes** (can be added over time – can be null)
- Maximum size of an item is **400KB**
- Data types supported are:
  - **Scalar Types** – String, Number, Binary, Boolean, Null
  - **Document Types** – List, Map
  - **Set Types** – String Set, Number Set, Binary Set

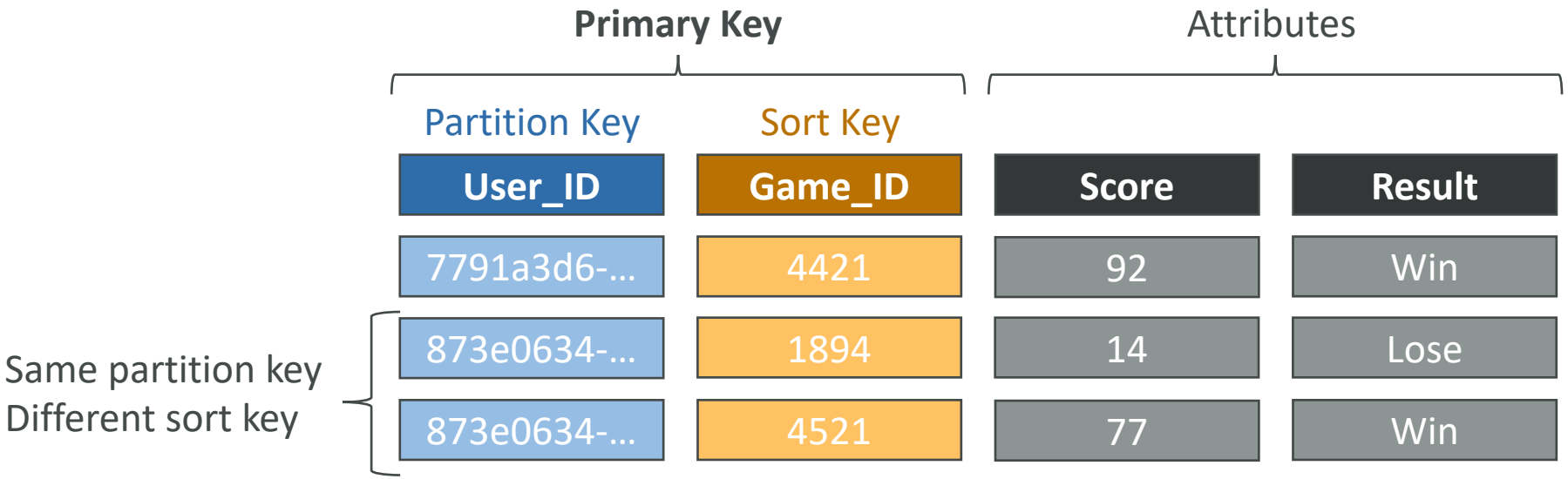
DynamoDB – Primary Keys

- Option 1: Partition Key (HASH)
  - Partition key must be unique for each item
  - Partition key must be "diverse" so that the data is distributed
  - Example: "User_ID" for a users table



DynamoDB – Primary Keys

- Option 2: Partition Key + Sort Key (HASH + RANGE)
  - The combination must be unique for each item
  - Data is grouped by partition key
  - Example: users-games table, "User_ID" for Partition Key and "Game_ID" for Sort Key



DynamoDB – Partition Keys (Exercise)

- We're building a movie database
- What is the best Partition Key to maximize data distribution?
  - movie_id
  - producer_name
  - leader_actor_name
  - movie_language

- "movie_id" has the highest cardinality so it's a good candidate
- "movie_language" doesn't take many values and may be skewed towards English so it's not a great choice for the Partition Key

DynamoDB – Read/Write Capacity Modes

- Control how you manage your table's capacity (read/write throughput)

- Provisioned Mode
  - You specify the number of reads/writes per second
  - You need to plan capacity beforehand
  - Pay for provisioned read & write capacity units

- On-Demand Mode (default)
  - Read/writes automatically scale up/down with your workloads
  - No capacity planning needed
  - Pay for what you use, more expensive ($$$)

- You can switch between different modes once every 24 hours

R/W Capacity Modes – Provisioned

- Table must have provisioned read and write capacity units
- **Read Capacity Units (RCU)** – throughput for reads
- **Write Capacity Units (WCU)** – throughput for writes
- Option to setup auto-scaling of throughput to meet demand
- Throughput can be exceeded temporarily using **"Burst Capacity"**
- If Burst Capacity has been consumed, you'll get a **"ProvisionedThroughputExceededException"**
- It's then advised to do an **exponential backoff** retry

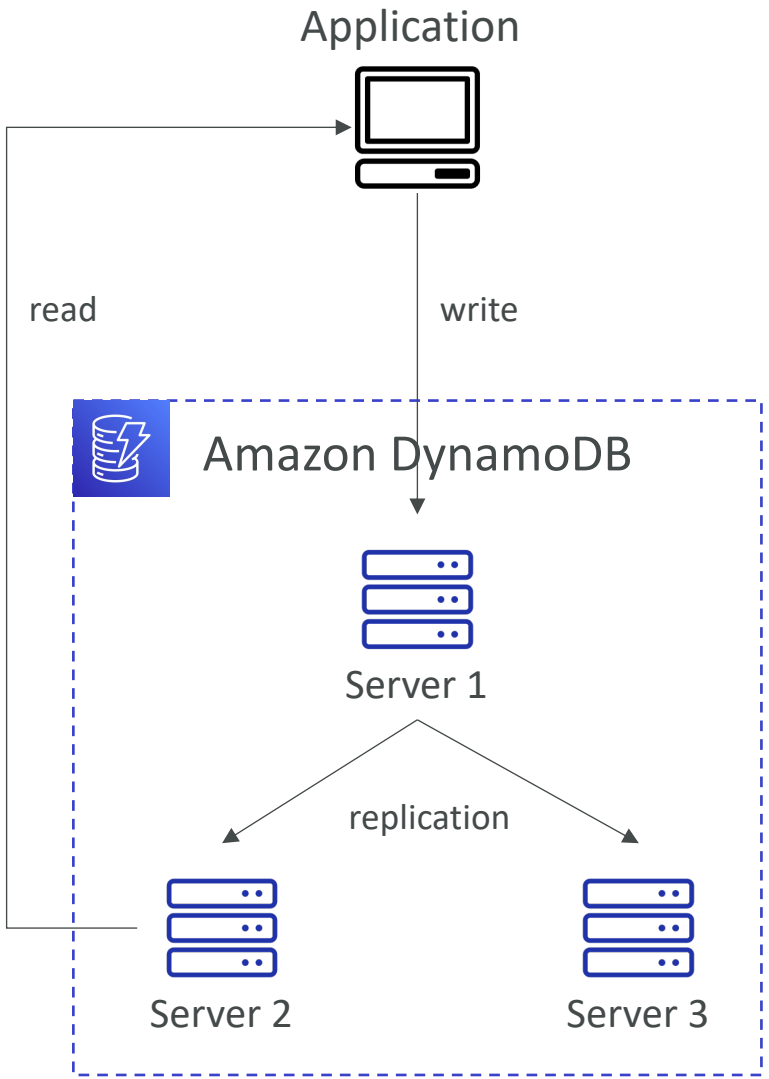
DynamoDB – Write Capacity Units (WCU)

- One *Write Capacity Unit (WCU)* represents one write per second for an item up to **1 KB** in size
- If the items are larger than 1 KB, more WCUs are consumed

- **Example 1:** we write 10 items per second, with item size 2 KB
  - We need $10 * (\frac{2\ KB}{1\ KB}) = 20\ WCUs$
- **Example 2:** we write 6 items per second, with item size 4.5 KB
  - We need $6 * (\frac{5\ KB}{1\ KB}) = 30\ WCUs$ (4.5 gets rounded to the upper KB)
- **Example 3:** we write 120 items per minute, with item size 2 KB
  - We need $\left(\frac{120}{60}\right) * (\frac{2\ KB}{1\ KB}) = 4\ WCUs$

Strongly Consistent Read vs. Eventually Consistent Read

- Eventually Consistent Read (default)
  - If we read just after a write, it's possible we'll get some stale data because of replication
- Strongly Consistent Read
  - If we read just after a write, we will get the correct data
  - Set "**ConsistentRead**" parameter to **True** in API calls (GetItem, BatchGetItem, Query, Scan)
  - Consumes twice the RCU



DynamoDB – Read Capacity Units (RCU)

- One *Read Capacity Unit (RCU)* represents **one Strongly Consistent Read** per second, or **two Eventually Consistent Reads** per second, for an item up to **4 KB** in size
- If the items are larger than 4 KB, more RCUs are consumed

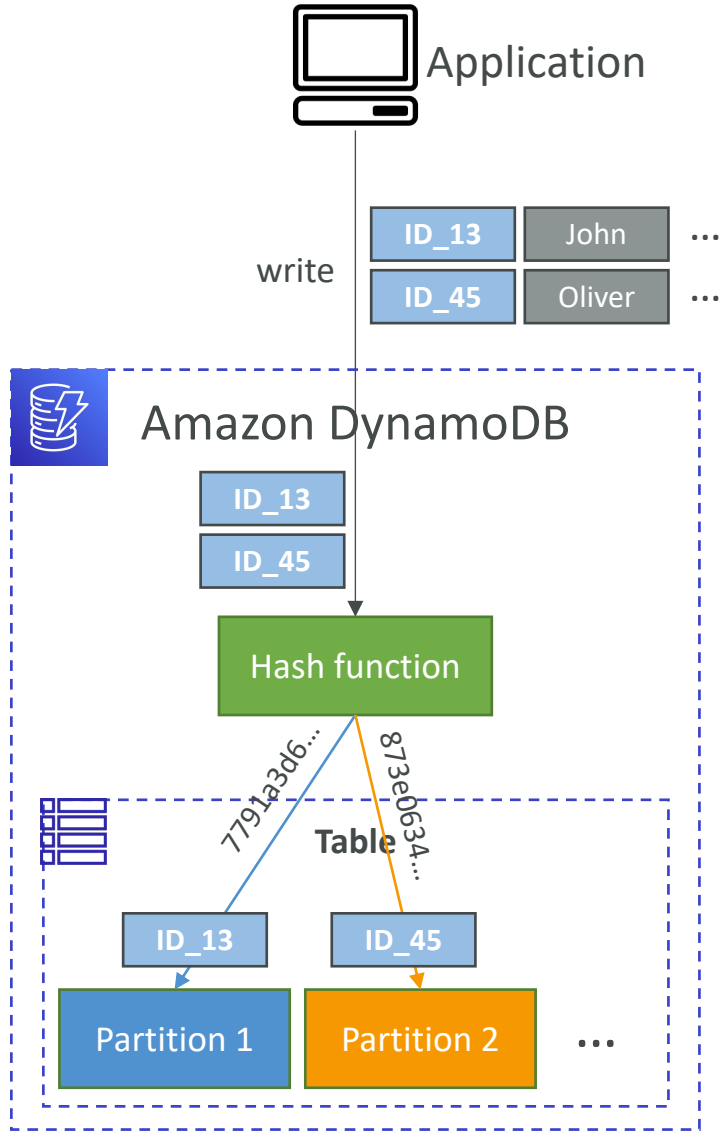
- **Example 1:** 10 Strongly Consistent Reads per second, with item size 4 KB
  - We need $10 * \left(\frac{4\ KB}{4\ KB}\right) = 10\ RCUs$
- **Example 2:** 16 Eventually Consistent Reads per second, with item size 12 KB
  - We need $\left(\frac{16}{2}\right) * \left(\frac{12\ KB}{4\ KB}\right) = 24\ RCUs$
- **Example 3:** 10 Strongly Consistent Reads per second, with item size 6 KB
  - We need $10 * \left(\frac{8\ KB}{4\ KB}\right) = 20\ RCUs$ (we must round up 6 KB to 8 KB)

DynamoDB – Partitions Internal

- Data is stored in partitions
- Partition Keys go through a hashing algorithm to know to which partition they go to

- To compute the number of partitions:
  - $\#\ of\ partitions_{by\ capacity} = \left(\frac{RCUs_{Total}}{3000}\right) + \left(\frac{WCUs_{Total}}{1000}\right)$
  - $\#\ of\ partitions_{by\ size} = \frac{Total\ Size}{10\ GB}$
  - $\#\ of\ partitions = \text{ceil}(\max(\#\ of\ partitions_{by\ capacity}, \#\ of\ partitions_{by\ size}))$

- WCUs and RCUs are spread evenly across partitions



DynamoDB – Throttling

- If we exceed provisioned RCUs or WCUs, we get "**ProvisionedThroughputExceededException**"
- Reasons:
  - **Hot Keys** – one partition key is being read too many times (e.g., popular item)
  - **Hot Partitions**
  - **Very large items,** remember RCU and WCU depends on size of items
- Solutions:
  - **Exponential backoff** when exception is encountered (already in SDK)
  - **Distribute partition keys** as much as possible
  - If RCU issue, we can **use DynamoDB Accelerator (DAX)**

R/W Capacity Modes – On-Demand

- Read/writes automatically scale up/down with your workloads
- No capacity planning needed (WCU / RCU)
- Unlimited WCU & RCU, no throttle, more expensive
- You're charged for reads/writes that you use in terms of **RRU** and **WRU**
- **Read Request Units (RRU)** – throughput for reads (same as RCU)
- **Write Request Units (WRU)** – throughput for writes (same as WCU)
- 2.5x more expensive than provisioned capacity (use with care)
- Use cases: unknown workloads, unpredictable application traffic, …

DynamoDB – Writing Data

- PutItem
  - Creates a new item or fully replace an old item (same Primary Key)
  - Consumes WCUs
- UpdateItem
  - Edits an existing item's attributes or adds a new item if it doesn't exist
  - Can be used to implement **Atomic Counters** – a numeric attribute that's unconditionally incremented
- Conditional Writes
  - Accept a write/update/delete only if conditions are met, otherwise returns an error
  - Helps with concurrent access to items
  - No performance impact

DynamoDB – Reading Data

- GetItem
  - Read based on Primary key
  - Primary Key can be HASH or HASH+RANGE
  - Eventually Consistent Read (default)
  - Option to use Strongly Consistent Reads (more RCU - might take longer)
  - ProjectionExpression can be specified to retrieve only certain attributes

DynamoDB – Reading Data (Query)

- **Query** returns items based on:
  - **KeyConditionExpression**
    - Partition Key value (**must be = operator**) – required
    - Sort Key value (=, <, <=, >, >=, Between, Begins with) – optional
  - **FilterExpression**
    - Additional filtering after the Query operation (before data returned to you)
    - Use only with non-key attributes (does not allow HASH or RANGE attributes)
- Returns:
  - The number of items specified in **Limit**
  - Or up to 1 MB of data
- Ability to do pagination on the results
- Can query table, a Local Secondary Index, or a Global Secondary Index

DynamoDB – Reading Data (Scan)

- **Scan** the entire table and then filter out data (inefficient)
- Returns up to 1 MB of data – use pagination to keep on reading
- Consumes a lot of RCU
- Limit impact using **Limit** or reduce the size of the result and pause
- For faster performance, use **Parallel Scan**
  - Multiple workers scan multiple data segments at the same time
  - Increases the throughput and RCU consumed
  - Limit the impact of parallel scans just like you would for Scans
- Can use **ProjectionExpression** & **FilterExpression** (no changes to RCU)

DynamoDB – Deleting Data

- DeleteItem
  - Delete an individual item
  - Ability to perform a conditional delete

- DeleteTable
  - Delete a whole table and all its items
  - Much quicker deletion than calling **DeleteItem** on all items

DynamoDB – Batch Operations

- Allows you to save in latency by reducing the number of API calls
- Operations are done in parallel for better efficiency
- Part of a batch can fail; in which case we need to try again for the failed items

- **BatchWriteItem**
  - Up to 25 **PutItem** and/or **DeleteItem** in one call
  - Up to 16 MB of data written, up to 400 KB of data per item
  - Can't update items (use **UpdateItem**)
  - **UnprocessedItems** for failed write operations (exponential backoff or add WCU)

- **BatchGetItem**
  - Return items from one or more tables
  - Up to 100 items, up to 16 MB of data
  - Items are retrieved in parallel to minimize latency
  - **UnprocessedKeys** for failed read operations (exponential backoff or add RCU)

DynamoDB – PartiQL

- SQL-compatible query language for DynamoDB
- Allows you to select, insert, update, and delete data in DynamoDB using SQL
- Run queries across multiple DynamoDB tables
- Run PartiQL queries from:
  - AWS Management Console
  - NoSQL Workbench for DynamoDB
  - DynamoDB APIs
  - AWS CLI
  - AWS SDK

```
SELECT OrderID, Total
FROM Orders
WHERE OrderID IN [1, 2, 3]
ORDER BY OrderID DESC
```

DynamoDB – Conditional Writes

- For **PutItem**, **UpdateItem**, **DeleteItem**, and **TransactWriteItems**
- You can specify a Condition expression to determine which items should be modified:
  - **attribute_exists**
  - **attribute_not_exists**
  - **attribute_type**
  - **contains** (for string)
  - **begins_with** (for string)
  - ProductCategory **IN** (:cat1, :cat2) **and** Price **between** :low **and** :high
  - **size** (string length)

- Note: Filter Expression filters the results of read queries, while Condition Expressions are for write operations

Conditional Writes – Example on Update Item

```
aws dynamodb update-item \
    --table-name ProductCatalog \
    --key '{ "Id": { "N": "456" } }' \
    --update-expression "SET Price = Price - :discount" \
    --condition-expression "Price > :limit" \
    --expression-attribute-values file://values.json
```

```
{
    ":discount": {
        "N": "150"
    },
    ":limit": {
        "N": "500"
    }
}
```

values.json

```
{
    "Id": {
        "N": "456"
    },
    "Price": {
        "N": "650"
    },
    "ProductCategory": {
        "S": "Sporting Goods"
    }
}
```

```
{
    "Id": {
        "N": "456"
    },
    "Price": {
        "N": "500"
    },
    "ProductCategory": {
        "S": "Sporting Goods"
    }
}
```

Conditional Writes – Example on Delete Item

- attribute_not_exists
  - Only succeeds if the attribute doesn't exist yet (no value)

```
aws dynamodb delete-item \
    --table-name ProductCatalog \
    --key '{ "Id": { "N": "456" } }' \
    --condition-expression "attribute_not_exists(Price)"
```

- attribute_exists
  - Opposite of attribute_not_exists

```
aws dynamodb delete-item \
    --table-name ProductCatalog \
    --key '{ "Id": { "N": "456" } }' \
    --condition-expression "attribute_exists(ProductReviews.OneStar)"
```

Conditional Writes – Do Not Overwrite Elements

- attribute_not_exists(partition_key)
  - Make sure the item isn't overwritten

- attribute_not_exists(partition_key) and attribute_not_exists(sort_key)
  - Make sure the partition / sort key combination is not overwritten

Conditional Writes – Example Complex Condition

```
aws dynamodb delete-item \
    --table-name ProductCatalog \
    --key '{ "Id": { "N": "456" } }' \
    --condition-expression "(ProductCategory IN (:cat1, :cat2)) and (Price between :lo and :hi)" \
    --expression-attribute-values file://values.json
```

```
{
    ":cat1": {
        "S": "Sporting Goods"
    },
    ":cat2": {
        "S": "Gardening Supplies"
    },
    ":lo": {
        "N": "500"
    },
    ":hi": {
        "N": "600"
    }
}
```

values.json

```
{
    "Id": {
        "N": "456"
    },
    "Price": {
        "N": "650"
    },
    "ProductCategory": {
        "S": "Sporting Goods"
    }
}
```

Conditional Writes – Example of String Comparisons

- **begins_with** – check if prefix matches
- **contains** – check if string is contained in another string

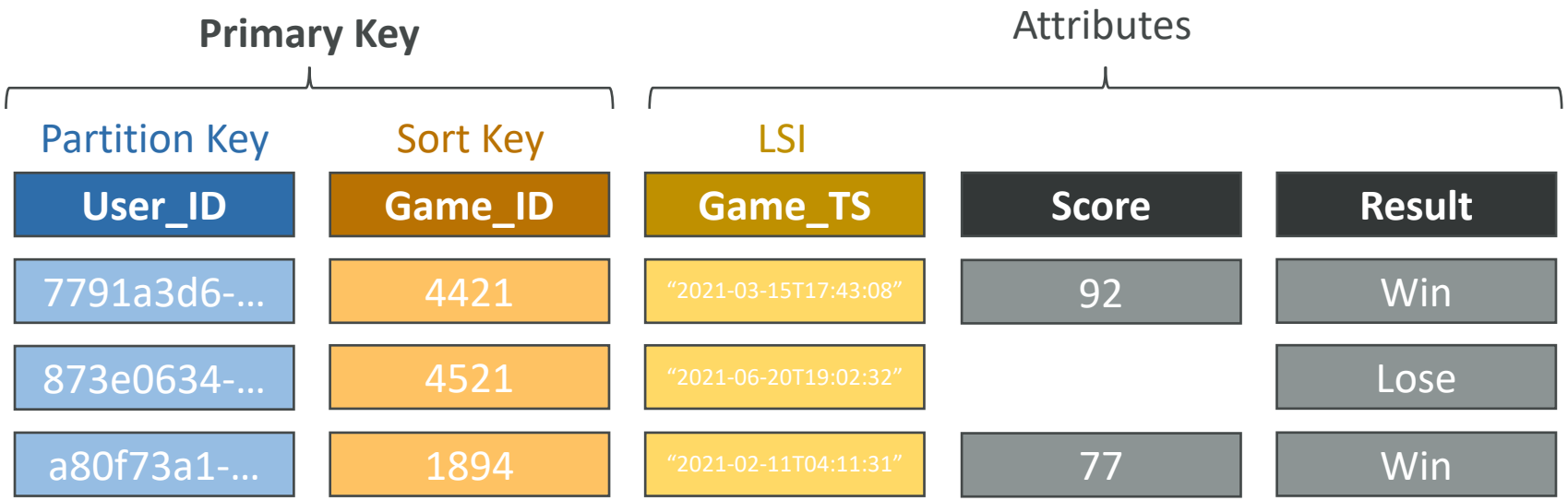
```
aws dynamodb delete-item \
    --table-name ProductCatalog \
    --key '{ "Id": { "N": "456" } }' \
    --condition-expression "begins_with(Pictures.FrontView, :v_sub)" \
    --expression-attribute-values file://values.json
```

```
{
    ":v_sub": {
        "S": "http://"
    }
}
```

values.json

DynamoDB – Local Secondary Index (LSI)

- **Alternative Sort Key** for your table (same **Partition Key** as that of base table)
- The Sort Key consists of one scalar attribute (String, Number, or Binary)
- Up to 5 Local Secondary Indexes per table
- **Must be defined at table creation time**
- **Attribute Projections** – can contain some or all the attributes of the base table (KEYS_ONLY, INCLUDE, ALL)



DynamoDB – Global Secondary Index (GSI)

- **Alternative Primary Key (HASH or HASH+RANGE)** from the base table
- Speed up queries on non-key attributes
- The Index Key consists of scalar attributes (String, Number, or Binary)
- **Attribute Projections** – some or all the attributes of the base table (**KEYS_ONLY**, **INCLUDE**, **ALL**)
- Must provision RCUs & WCUs for the index
- **Can be added/modified after table creation**

Partition Key	Sort Key	Attributes
User_ID	**Game_ID**	**Game_TS**
7791a3d6-...	4421	"2021-03-15T17:43:08"
873e0634-...	4521	"2021-06-20T19:02:32"
a80f73a1-...	1894	"2021-02-11T04:11:31"

TABLE (query by "User_ID")

Partition Key	Sort Key	Attributes
Game_ID	**Game_TS**	**User_ID**
4421	"2021-03-15T17:43:08"	7791a3d6-...
4521	"2021-06-20T19:02:32"	873e0634-...
1894	"2021-02-11T04:11:31"	a80f73a1-...

INDEX GSI (query by "Game_ID")

DynamoDB – Indexes and Throttling

- Global Secondary Index (GSI):
  - If the writes are throttled on the GSI, then the main table will be throttled!
  - Even if the WCU on the main tables are fine
  - Choose your GSI partition key carefully!
  - Assign your WCU capacity carefully!

- Local Secondary Index (LSI):
  - Uses the WCUs and RCUs of the main table
  - No special throttling considerations

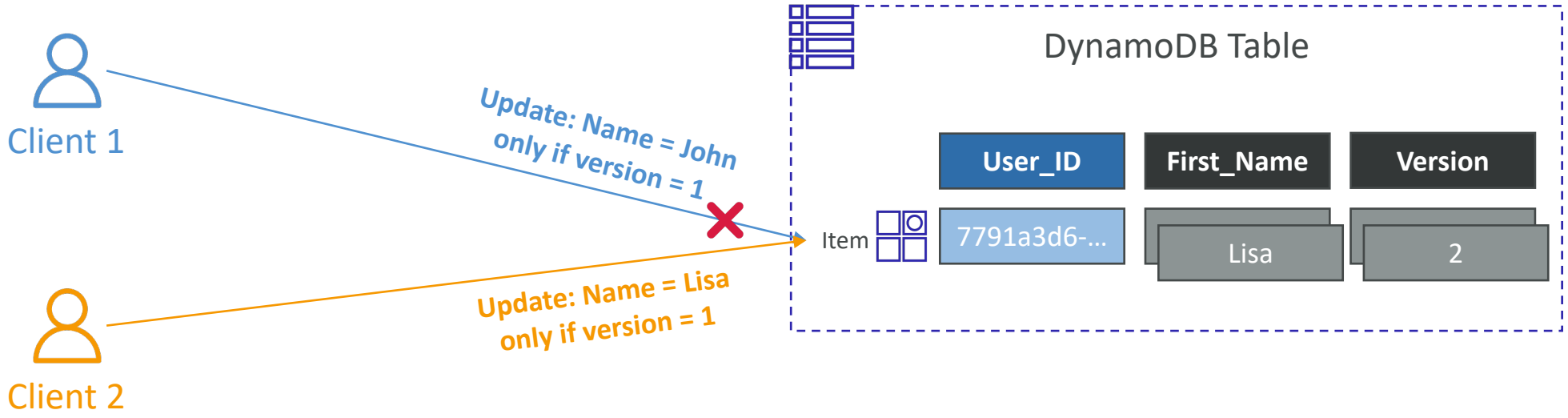
DynamoDB - PartiQL

- Use a SQL-like syntax to manipulate DynamoDB tables

- Supports some (but not all) statements:
  - INSERT
  - UPDATE
  - SELECT
  - DELETE
- It supports Batch operations

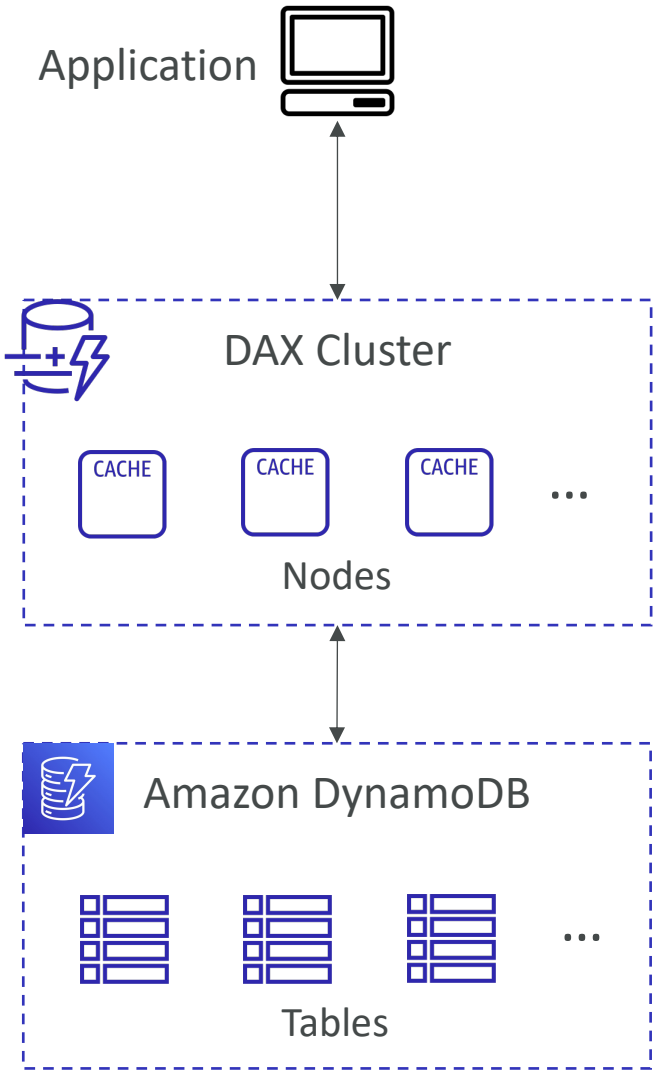
DynamoDB – Optimistic Locking

- DynamoDB has a feature called "**Conditional Writes**"
- A strategy to ensure an item hasn't changed before you update/delete it
- Each item has an attribute that acts as a *version number*

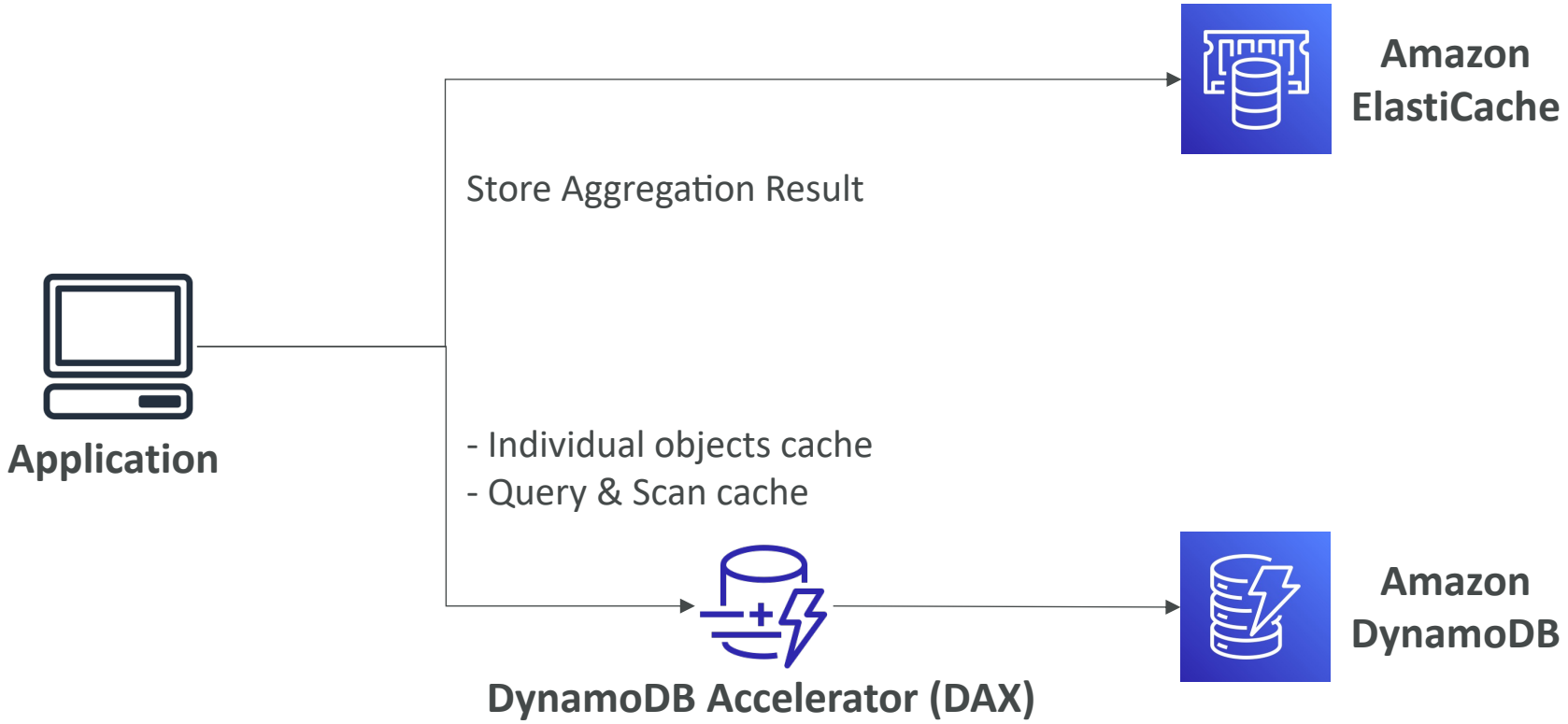


DynamoDB Accelerator (DAX)

- Fully-managed, highly available, seamless in-memory cache for DynamoDB
- Microseconds latency for cached reads & queries
- Doesn't require application logic modification (compatible with existing DynamoDB APIs)
- Solves the "Hot Key" problem (too many reads)
- 5 minutes TTL for cache (default)
- Up to 10 nodes in the cluster
- Multi-AZ (3 nodes minimum recommended for production)
- Secure (Encryption at rest with KMS, VPC, IAM, CloudTrail, …)



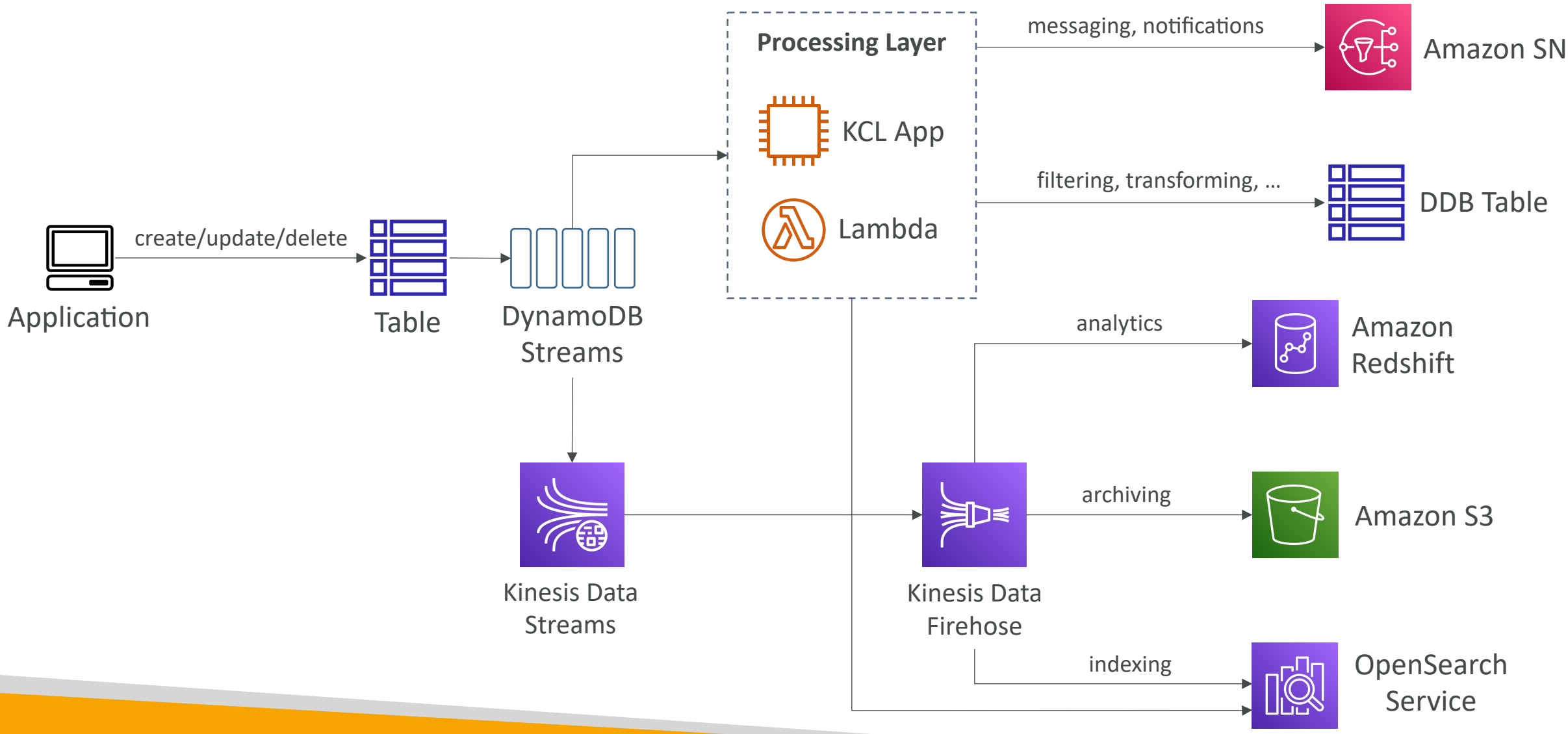
DynamoDB Accelerator (DAX) vs. ElastiCache



DynamoDB Streams

- Ordered stream of item-level modifications (create/update/delete) in a table
- Stream records can be:
  - Sent to **Kinesis Data Streams**
  - Read by **AWS Lambda**
  - Read by **Kinesis Client Library applications**
- Data Retention for up to 24 hours
- Use cases:
  - react to changes in real-time (welcome email to users)
  - Analytics
  - Insert into derivative tables
  - Insert into OpenSearch Service
  - Implement cross-region replication

DynamoDB Streams



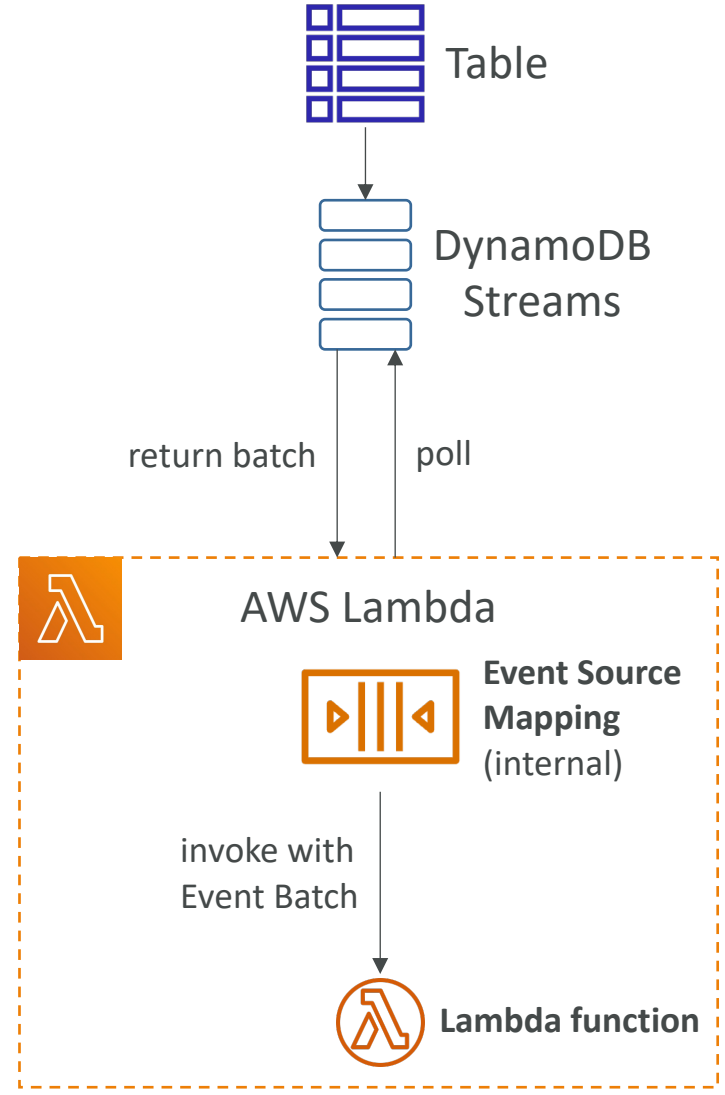
DynamoDB Streams

- Ability to choose the information that will be written to the stream:
  - KEYS_ONLY – only the key attributes of the modified item
  - NEW_IMAGE – the entire item, as it appears after it was modified
  - OLD_IMAGE – the entire item, as it appeared before it was modified
  - NEW_AND_OLD_IMAGES – both the new and the old images of the item
- DynamoDB Streams are made of shards, just like Kinesis Data Streams
- You don't provision shards, this is automated by AWS

- Records are not retroactively populated in a stream after enabling it

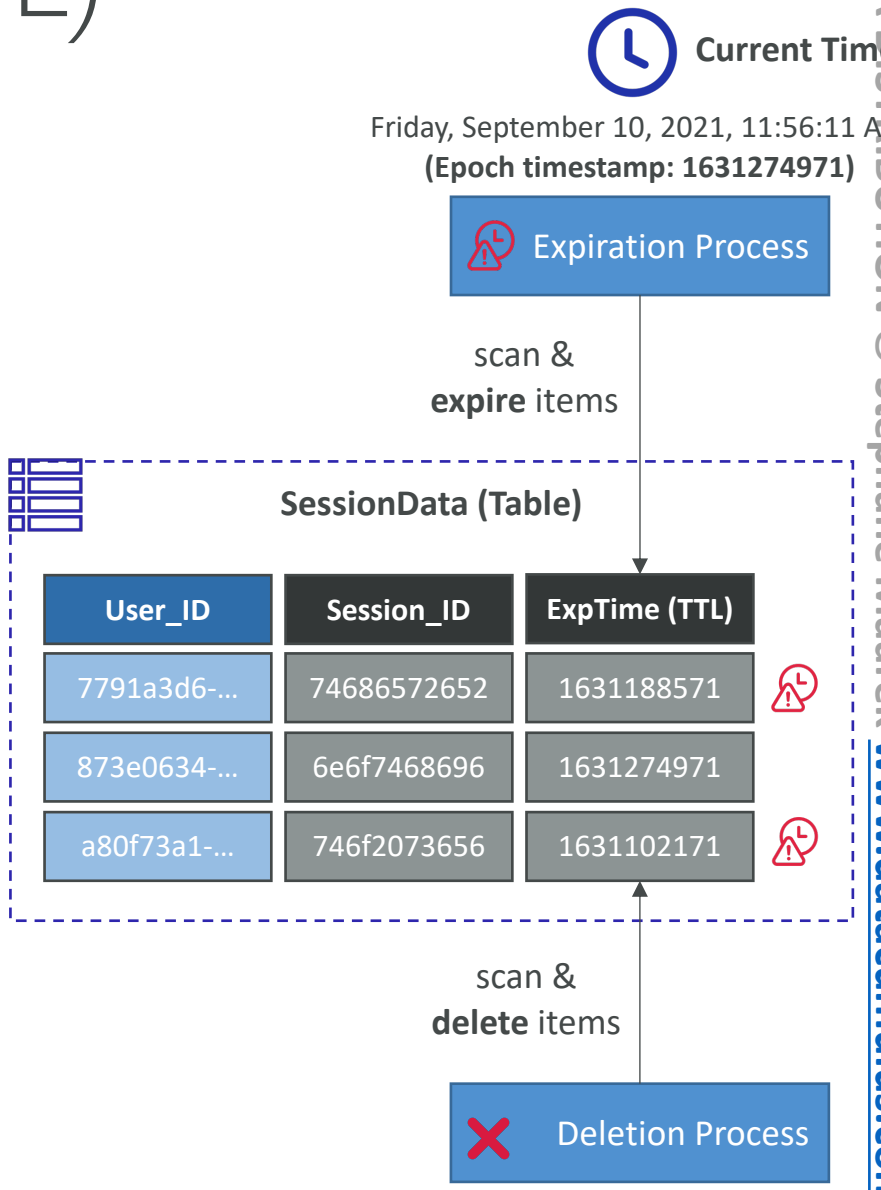
DynamoDB Streams & AWS Lambda

- You need to define an **Event Source Mapping** to read from a DynamoDB Streams
- You need to ensure the Lambda function has the **appropriate permissions**
- Your Lambda function is invoked synchronously



DynamoDB – Time To Live (TTL)

- Automatically delete items after an expiry timestamp
- Doesn't consume any WCUs (i.e., no extra cost)
- The TTL attribute must be a **"Number"** data type with **"Unix Epoch timestamp"** value
- Expired items deleted within few days of expiration
- Expired items, that haven't been deleted, appears in reads/queries/scans (if you don't want them, filter them out)
- Expired items are deleted from both LSIs and GSIs
- A delete operation for each expired item enters the DynamoDB Streams (can help recover expired items)
- Use cases: reduce stored data by keeping only current items, adhere to regulatory obligations, …



DynamoDB CLI – Good to Know

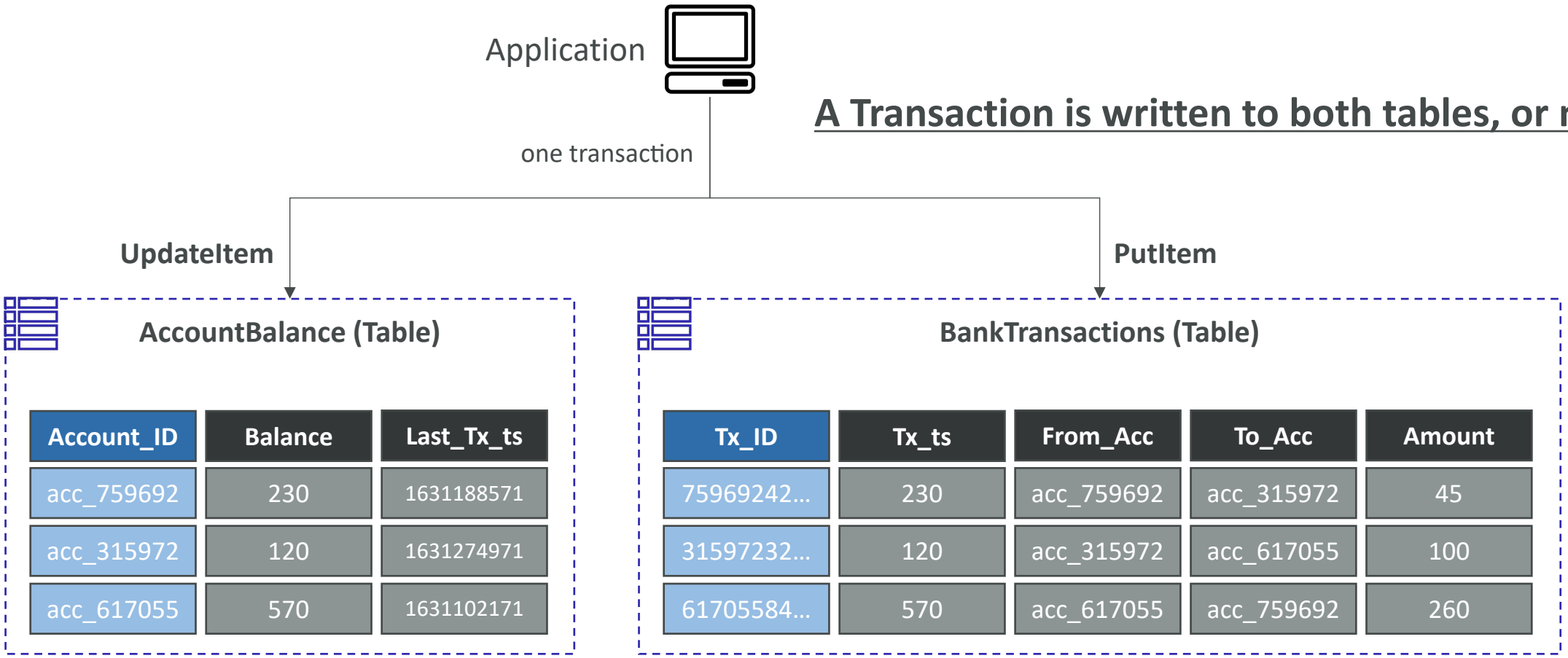
- **--projection-expression:** one or more attributes to retrieve
- **--filter-expression:** filter items before returned to you

- General AWS CLI Pagination options (e.g., DynamoDB, S3, …)
  - **--page-size:** specify that AWS CLI retrieves the full list of items but with a larger number of API calls instead of one API call (default: 1000 items)
  - **--max-items:** max. number of items to show in the CLI (returns **NextToken**)
  - **--starting-token:** specify the last **NextToken** to retrieve the next set of items

DynamoDB Transactions

- Coordinated, all-or-nothing operations (add/update/delete) to multiple items across one or more tables
- Provides Atomicity, Consistency, Isolation, and Durability (ACID)
- **Read Modes** – Eventual Consistency, Strong Consistency, Transactional
- **Write Modes** – Standard, Transactional
- **Consumes 2x WCUs & RCUs**
  - DynamoDB performs 2 operations for every item (prepare & commit)
- Two operations:
  - **TransactGetItems** – one or more **GetItem** operations
  - **TransactWriteItems** – one or more **PutItem**, **UpdateItem**, and **DeleteItem** operations
- Use cases: financial transactions, managing orders, multiplayer games, …

DynamoDB Transactions



DynamoDB Transactions – Capacity Computations

- Important for the exam!

- **Example 1:** 3 Transactional writes per second, with item size 5 KB
  - We need $3 * \left(\frac{5\,KB}{1\,KB}\right) * 2\ (transactional\ cost) = 30\ WCUs$

- **Example 2:** 5 Transaction reads per second , with item size 5 KB
  - We need $5 * \left(\frac{8\,KB}{4\,KB}\right) * 2\ (transactional\ cost) = 20\ RCUs$
  - (5 gets rounded to the upper 4 KB)

DynamoDB as Session State Cache

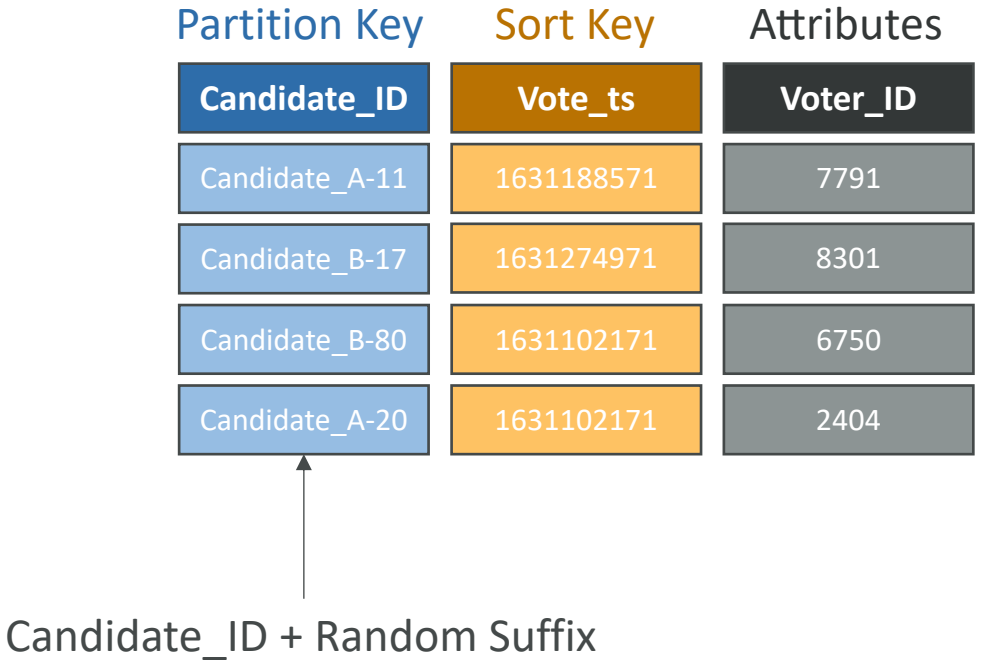
- It's common to use DynamoDB to store session states

- vs. ElastiCache
  - ElastiCache is in-memory, but DynamoDB is serverless
  - Both are key/value stores
- vs. EFS
  - EFS must be attached to EC2 instances as a network drive
- vs. EBS & Instance Store
  - EBS & Instance Store can only be used for local caching, not shared caching
- vs. S3
  - S3 is higher latency, and not meant for small objects

DynamoDB Write Sharding

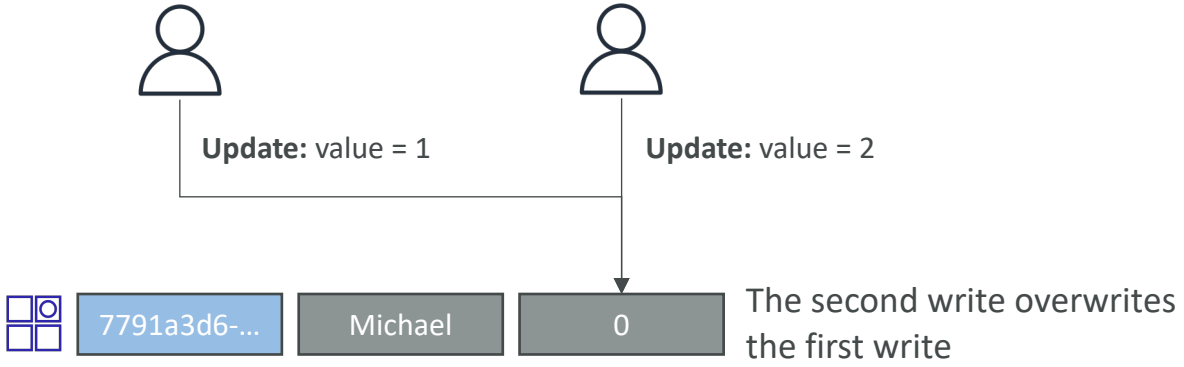
- Imagine we have a voting application with two candidates, **candidate A** and **candidate B**
- If **Partition Key** is **"Candidate_ID"**, this results into two partitions, which will generate issues (e.g., Hot Partition)

- A strategy that allows better distribution of items evenly across partitions
- **Add a suffix to Partition Key value**
- Two methods:
  - Sharding Using Random Suffix
  - Sharding Using Calculated Suffix



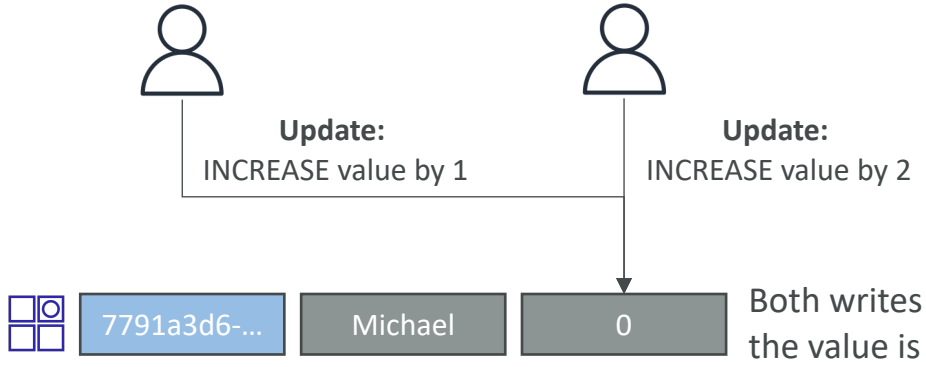
DynamoDB – Write Types

Concurrent Writes



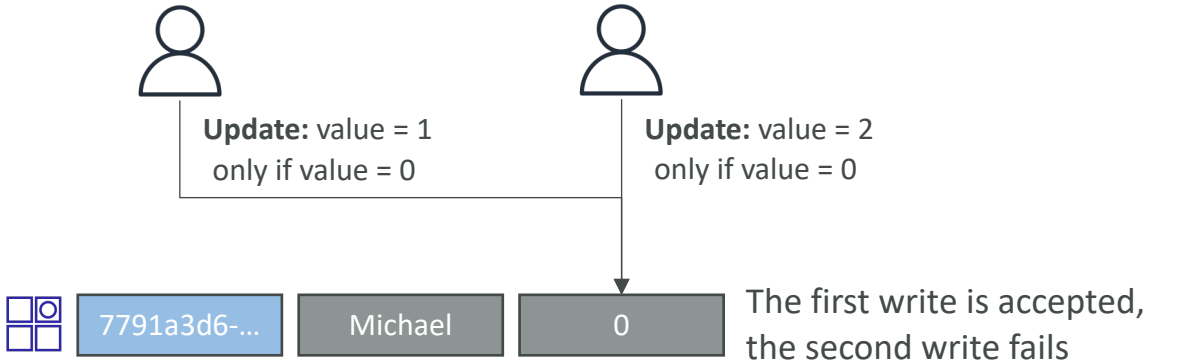
The second write overwrites the first write

Atomic Writes



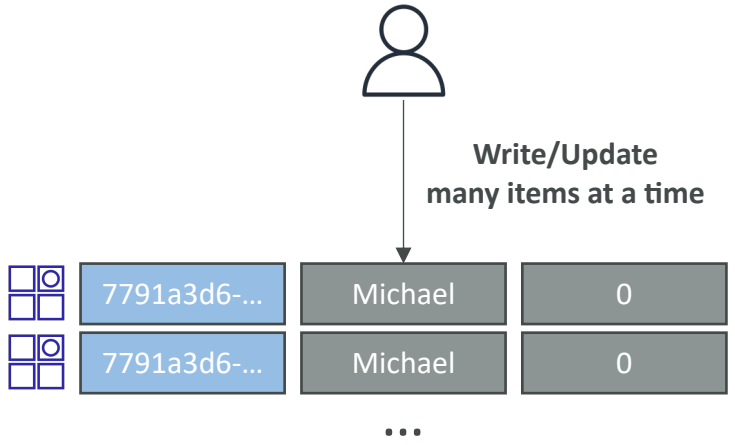
Both writes succeed, the value is increased by 3

Conditional Writes

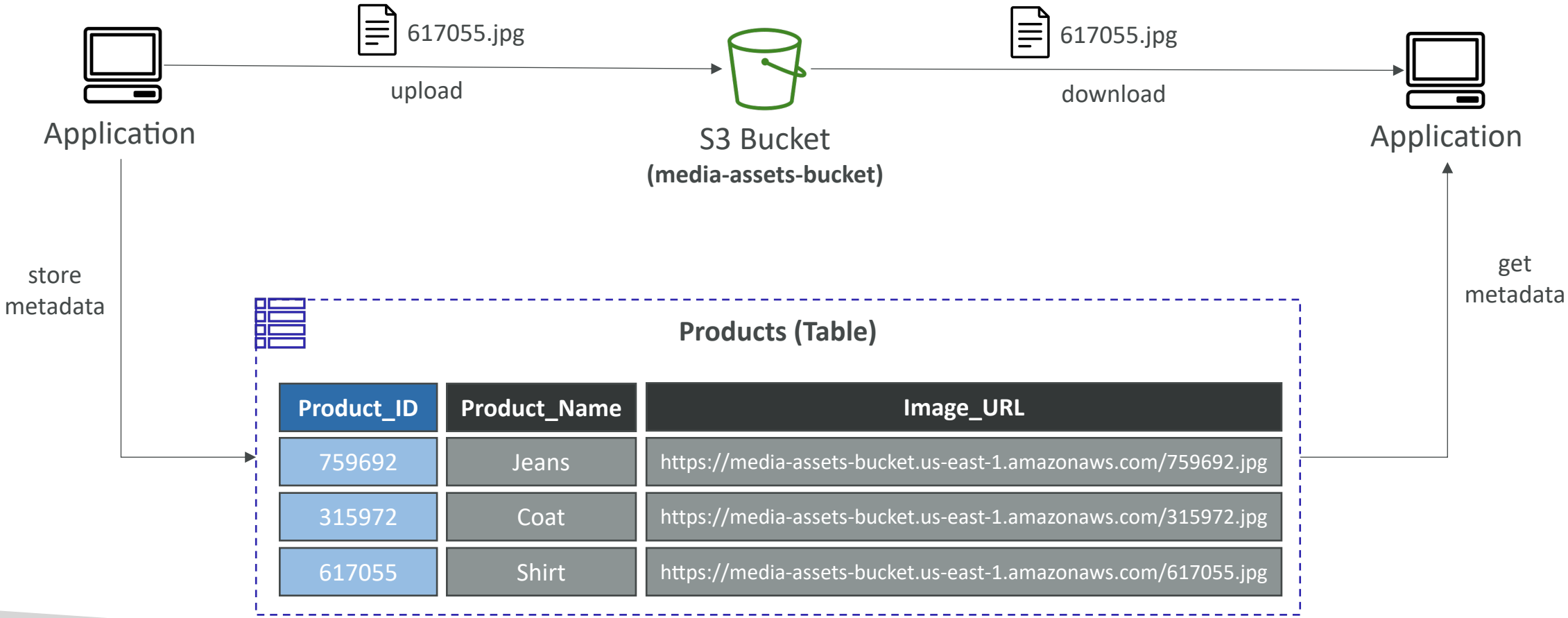


The first write is accepted, the second write fails

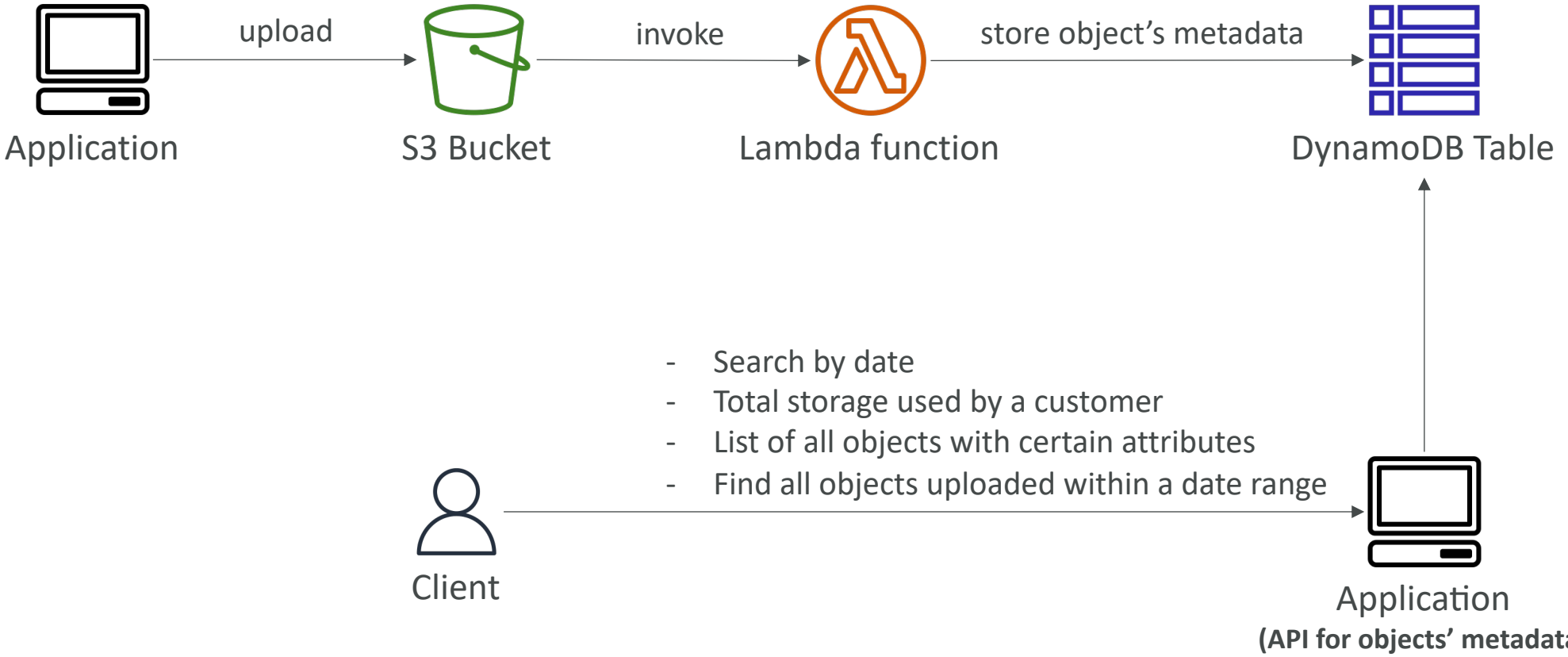
Batch Writes



DynamoDB – Large Objects Pattern



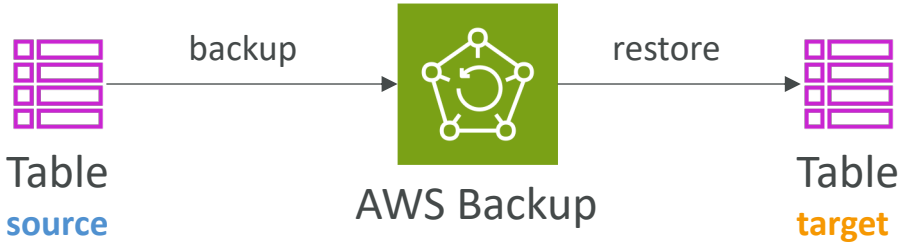
DynamoDB – Indexing S3 Objects Metadata



DynamoDB Operations

- Table Cleanup
  - Option 1: Scan + DeleteItem
    - Very slow, consumes RCU & WCU, expensive
  - Option 2: Drop Table + Recreate table
    - Fast, efficient, cheap

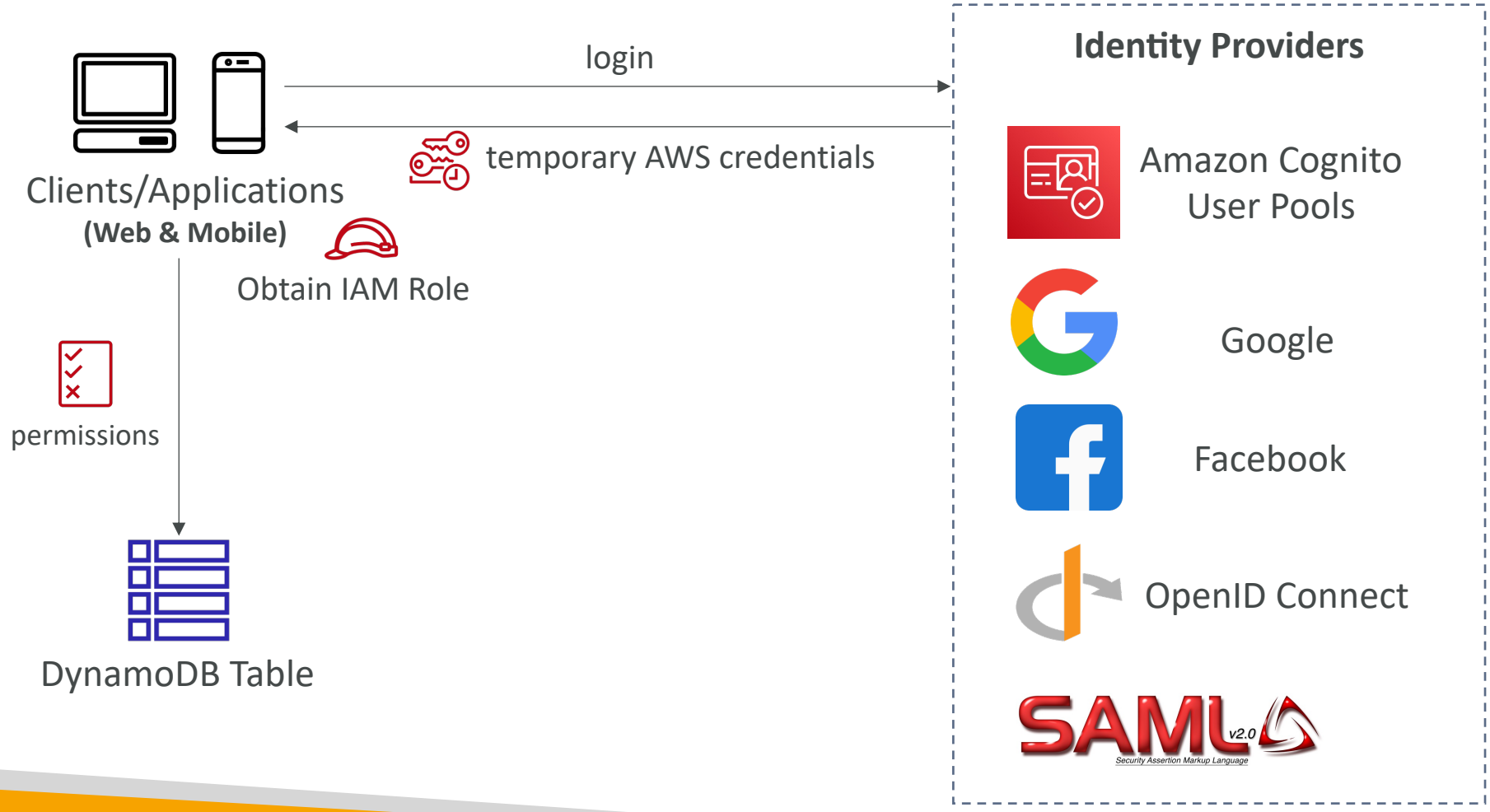
- Copying a DynamoDB Table
  - Option 1: Using AWS Backup
  - Option 2: Using AWS Glue
  - Option 3: Scan + PutItem or BatchWriteItem
    - Write your own code



DynamoDB – Security & Other Features

- Security
  - VPC Endpoints available to access DynamoDB without using the Internet
  - Access fully controlled by IAM
  - Encryption at rest using AWS KMS and in-transit using SSL/TLS
- Backup and Restore feature available
  - Point-in-time Recovery (PITR) like RDS
  - No performance impact
- Global Tables
  - Multi-region, multi-active, fully replicated, high performance
- DynamoDB Local
  - Develop and test apps locally without accessing the DynamoDB web service (without Internet)
- AWS Database Migration Service (AWS DMS) can be used to migrate to DynamoDB (from MongoDB, Oracle, MySQL, S3, …)

DynamoDB – Users Interact with DynamoDB Directly



DynamoDB – Fine-Grained Access Control

- Using **Web Identity Federation** or **Cognito Identity Pools**, each user gets AWS credentials
- You can assign an IAM Role to these users with a **Condition** to limit their API access to DynamoDB
- **LeadingKeys** – limit row-level access for users on the Primary Key
- **Attributes** – limit specific attributes the user can see

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "dynamodb:GetIem", "dynamodb:BatchGetItem", "dynamodb:Query",
                "dynamodb:PutItem", "dynamodb:UpdateItem", "dynamodb:DeleteItem",
                "dynamodb:BatchWriteItem"
            ],
            "Resource": "arn:aws:dynamodb:us-west-2:123456789012:table/MyTable",
            "Condition": {
                "ForAllValues:StringEquals": {
                    "dynamodb:LeadingKeys": ["${cognito-identity.amazonaws.com:sub}"]
                }
            }
        }
    ]
}
```

More at: https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/specifying-conditions.html